

In Class Discussion

EN 601.124: The Ethics of Artificial Intelligence and Automation

Department of Computer Science, February 25th 2026





What is a word embedding, and how does the paper use it to measure stereotypes? How would you explain to a friend without technical jargon, i.e., just explain the basic idea as if you were telling a friend



Imagine you could run a word embedding analysis on the text messages, tweets, and search histories of your own social circle over the last five years. What stereotypes do you think it would surface?



Both papers study bias through very different methods — one analyzes text data over 100 years, the other runs a controlled lab experiment. What can each method tell us that the other can't? If you had to combine their findings into one argument about how stereotypes work, what would it be?



We talked in the class about stereotypes and how they show up in the language of recommendation letters, so that, for instance, women are described as "organized" or "warm". What if female candidates truly are more organized or warm? How does that impact the validity of the claim that language reproduces stereotypes?



Both papers have implications for how we might intervene on bias — whether in AI systems or in people. Based on the two readings, where do you think intervention is more feasible: changing the data that trains language models, or changing the learning environments that shape human prejudice?



Both papers use very different kinds of evidence. Rank the following from most to least convincing as proof that stereotypes cause real-world harm, and be ready to explain your order:

- 1. Word embedding patterns in historical text**
- 2. A controlled lab learning task**
- 3. Self-reported prejudice scores on a survey**
- 4. Real-world hiring or lending data showing disparate outcomes**
- 5. Brain imaging data showing differential neural responses to different groups**
- 6. Audit studies where identical resumes with different names get different callback rates**
- 7. Interviews with people who experienced discrimination**